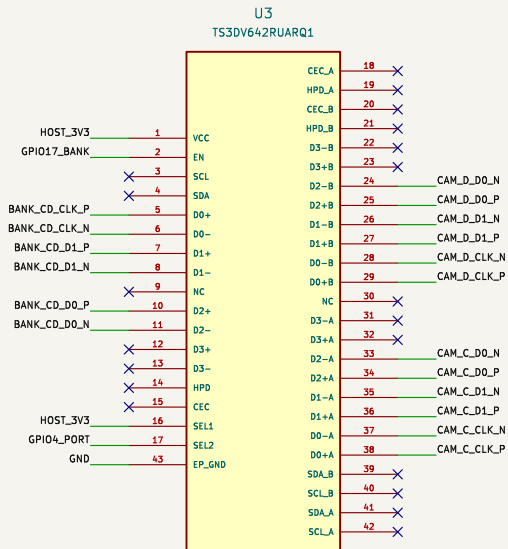
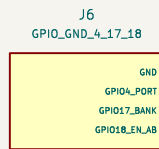
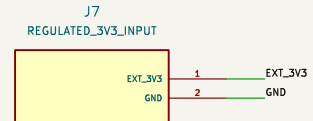
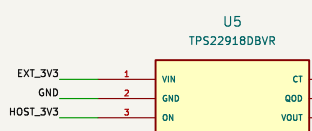
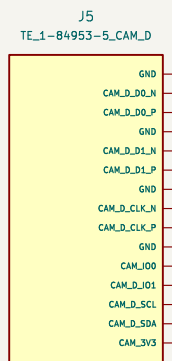
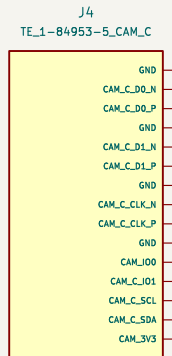
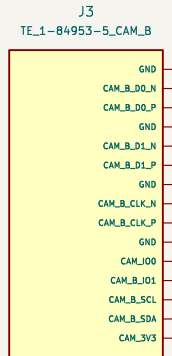
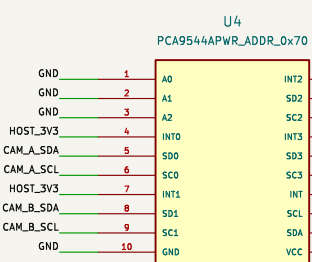
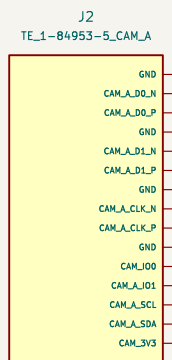
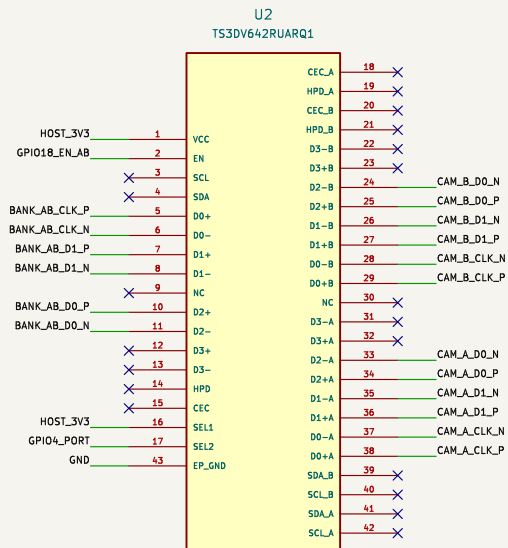
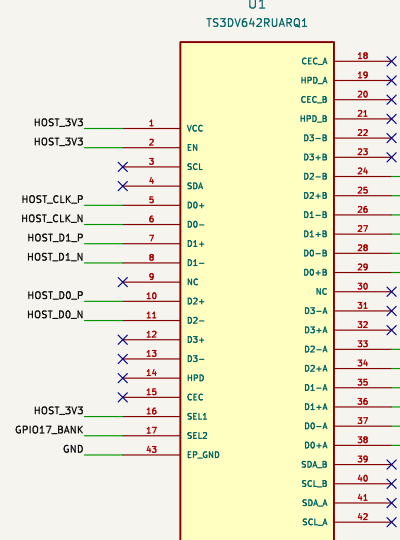
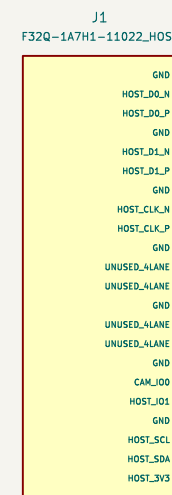
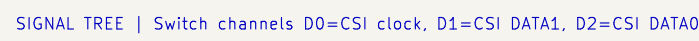
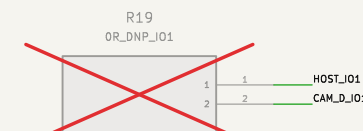
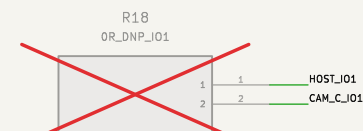
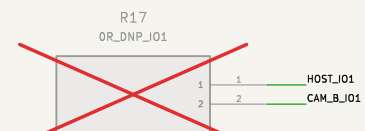
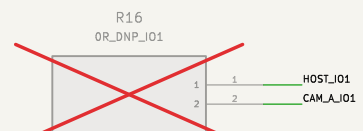
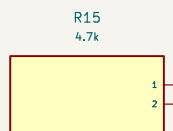
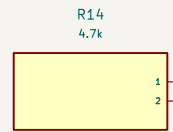
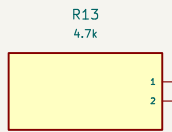
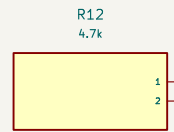
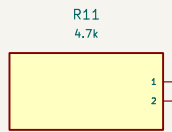
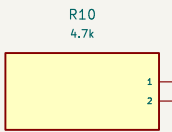
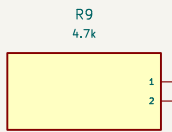
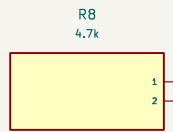
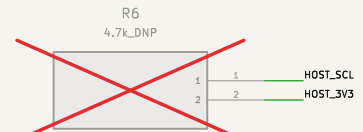
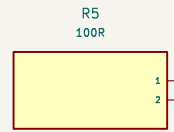
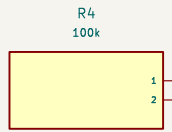
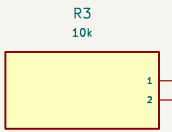
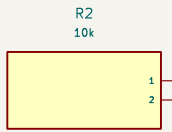
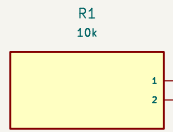
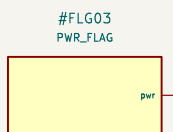
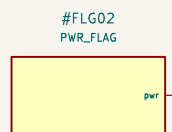
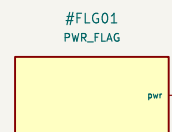
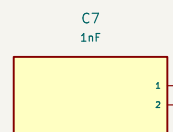
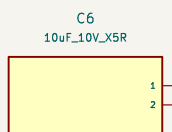
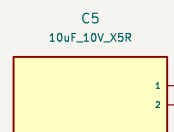
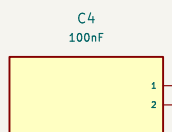
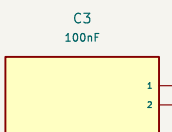
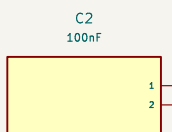
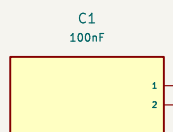




DECOUPLING AND CONTROL | 10k GPIO pulls disable both leaf switches at reset



A: GPIO4/18/17 = 0/1/0; B = 1/1/0; C = 0/0/1; D = 1/0/1

Camera I00 is common for probing. I01 links R16–R19 are DNP pending camera-specific review.

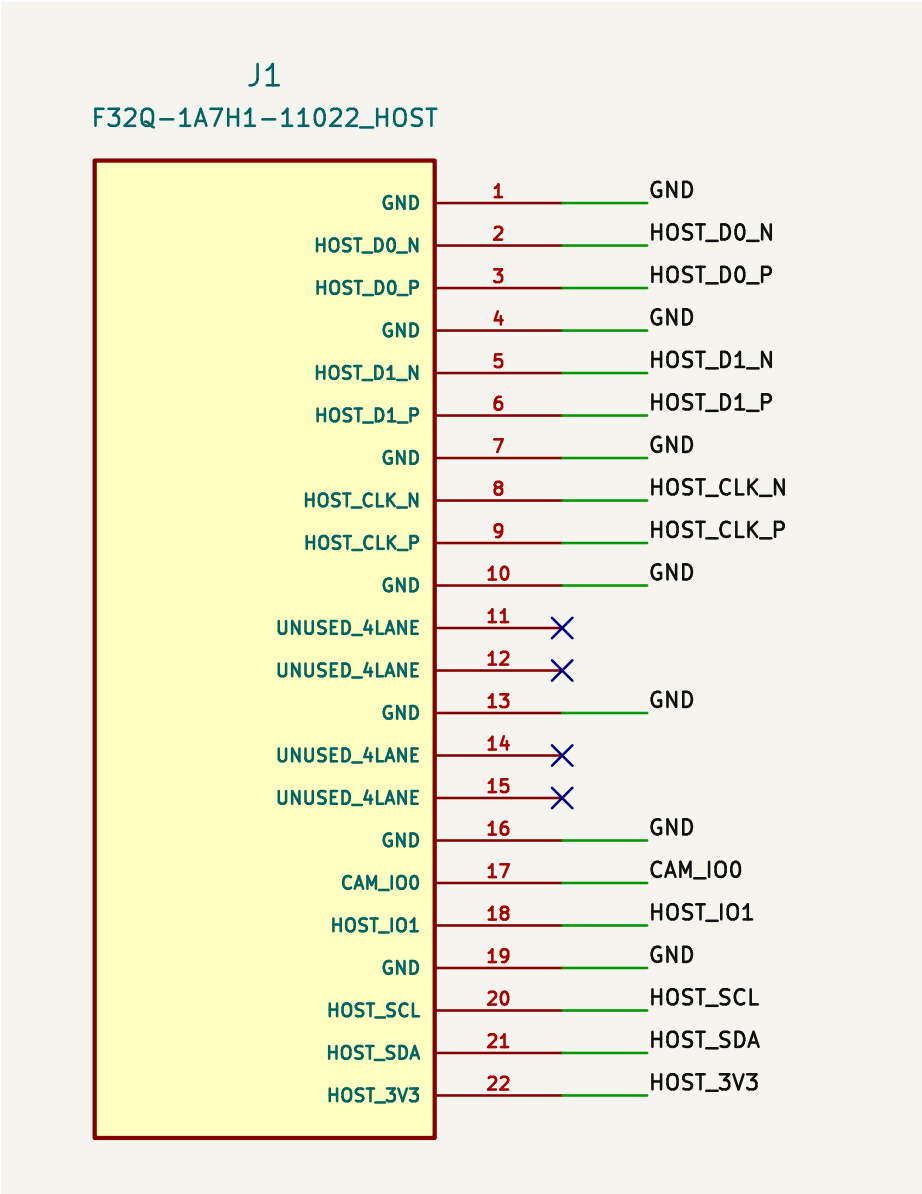
R6/R7 DNP: host pull-ups normally present. Check parallel module pull-ups before fitting R8-R15.

Not fabrication released: validate routing, SI, power and software
Two CSI data lanes; DNE selected camera; separate 3.3V camera supply
Simon hardware example
Sheet: /
File: pi5-camera-mux.kicad_sch
Title: Pi 5 / four-camera CSI selector – engineering prototype
Size: A1 Date: 2026-09-30 Rev: 0.1
KiCad E.D.A. 9.0.2 Id: 1/1

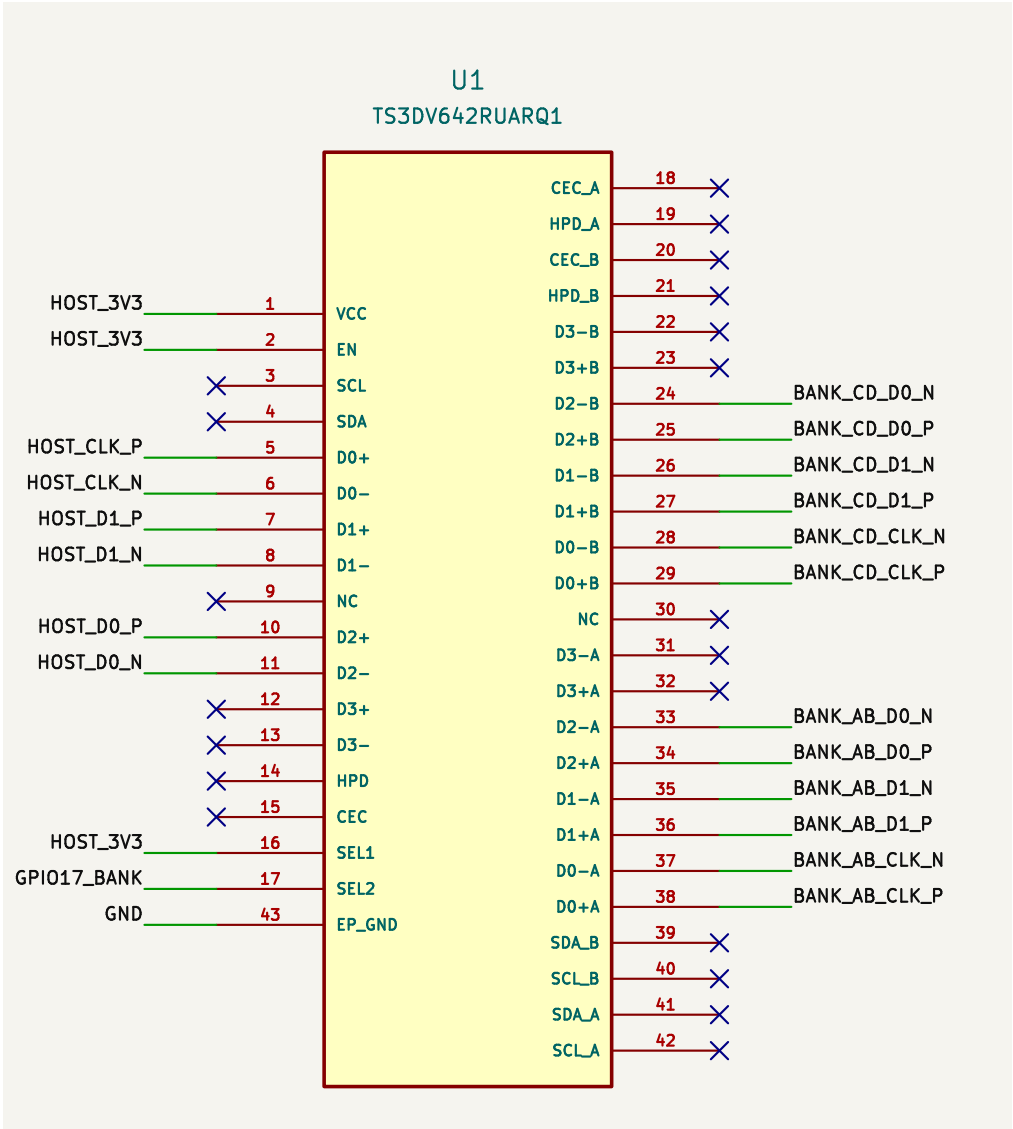
Signal path / host and root switch

J1 host connector and U1 bank selector

J1 / Raspberry Pi 5 host



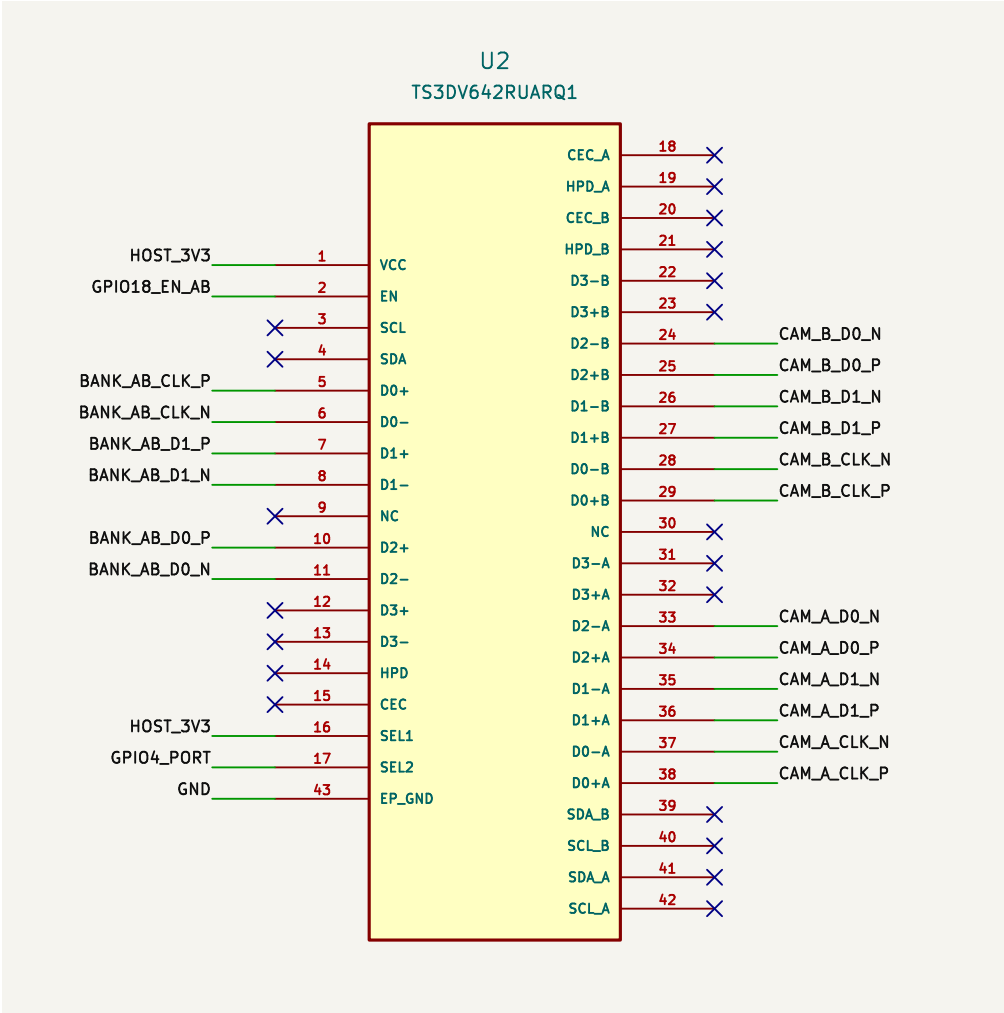
U1 / root switch



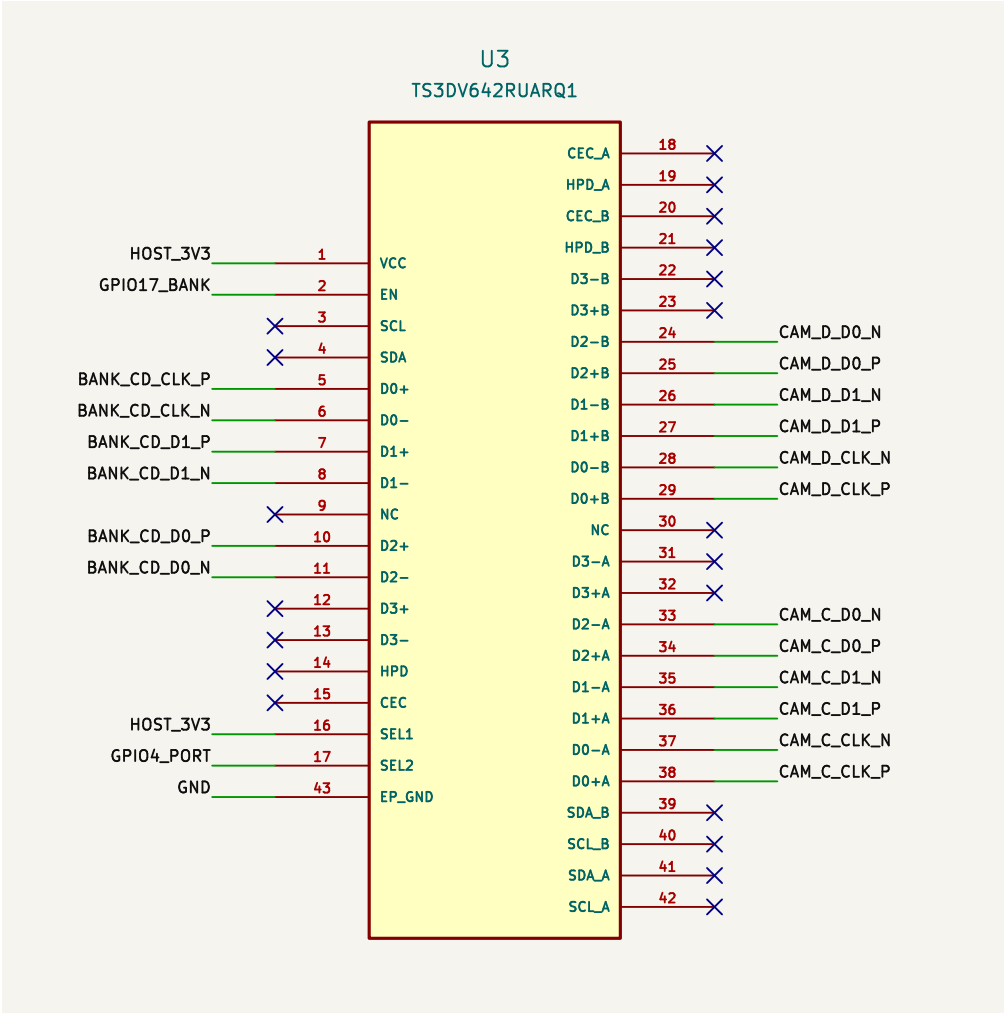
Signal path / leaf switches

Bank A/B and bank C/D; original net labels retained

U2 / cameras A and B



U3 / cameras C and D

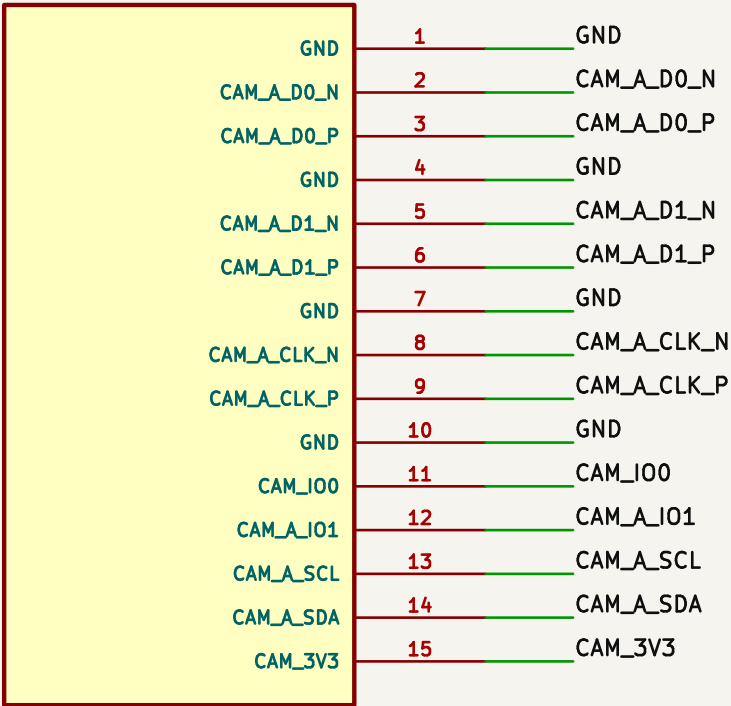


Camera connectors / A and B

15-position camera connectors; original pin numbering and IO labels

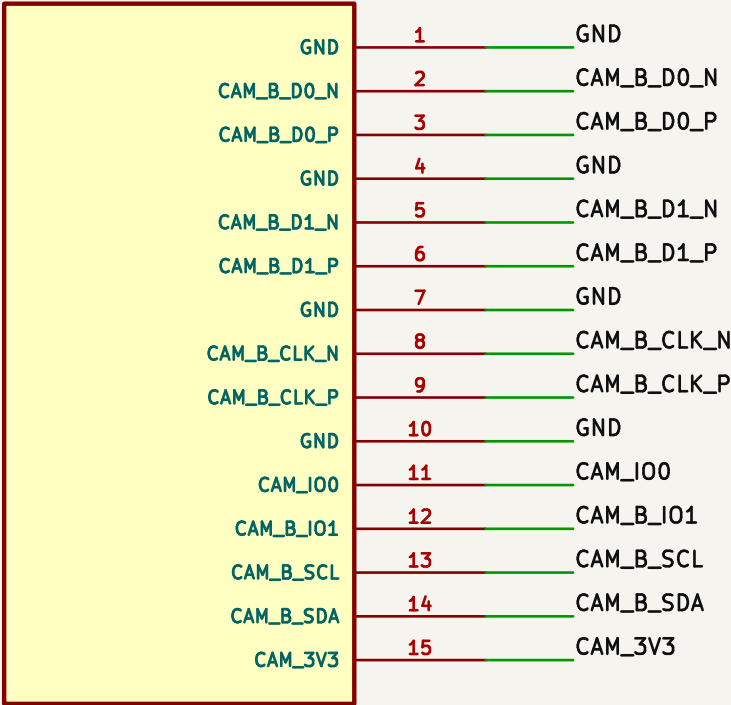
J2

J2
TE_1-84953-5_CAM_A



J3

J3
TE_1-84953-5_CAM_B

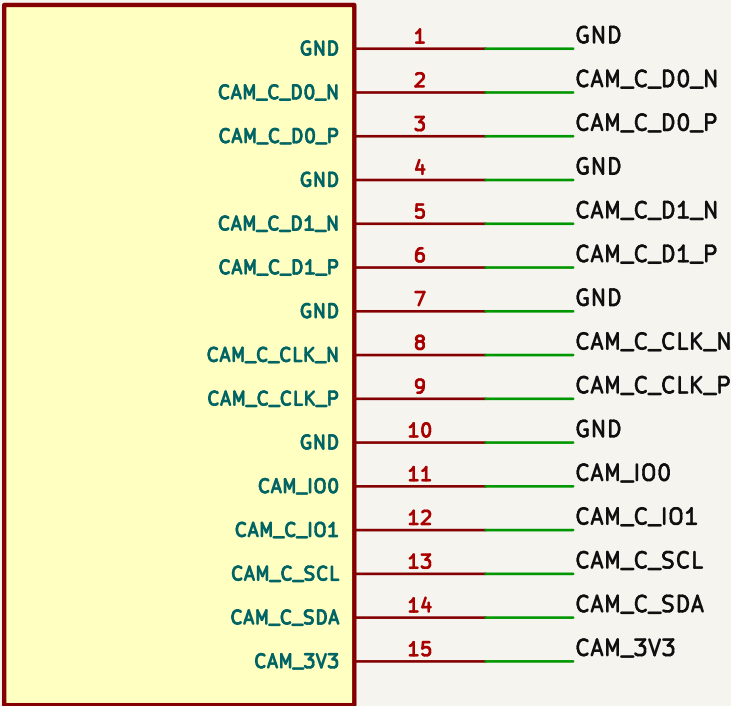


Camera connectors / C and D

15-position camera connectors; original pin numbering and IO labels

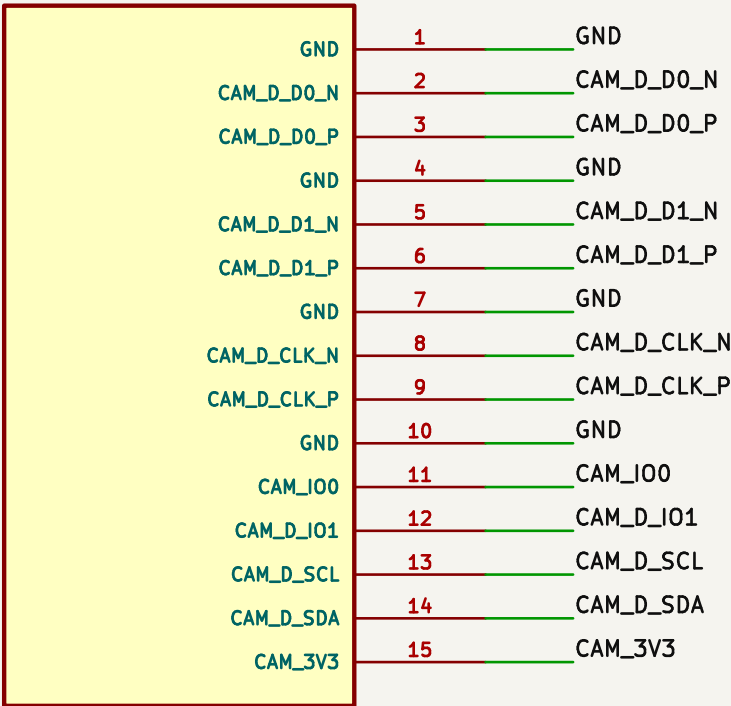
J4

J4
TE_1-84953-5_CAM_C



J5

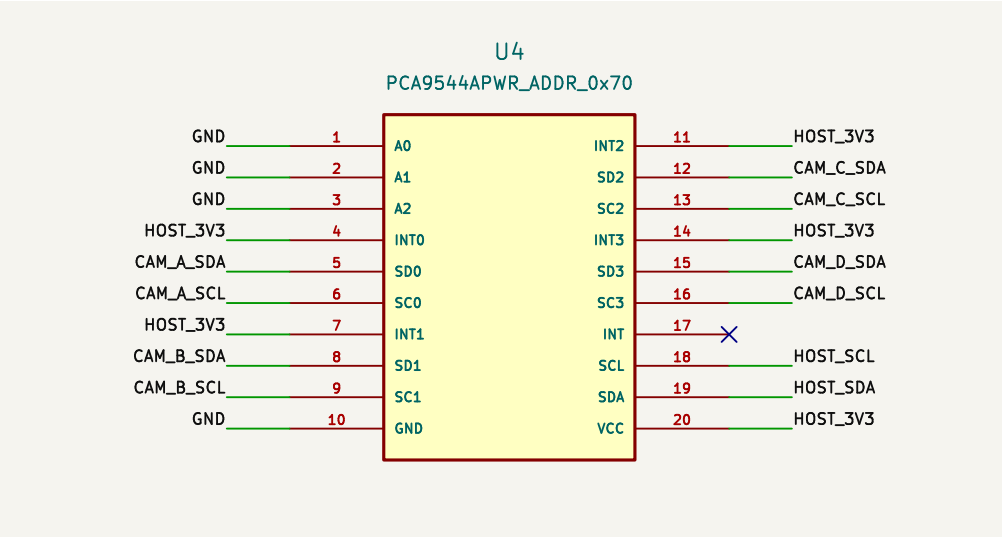
J5
TE_1-84953-5_CAM_D



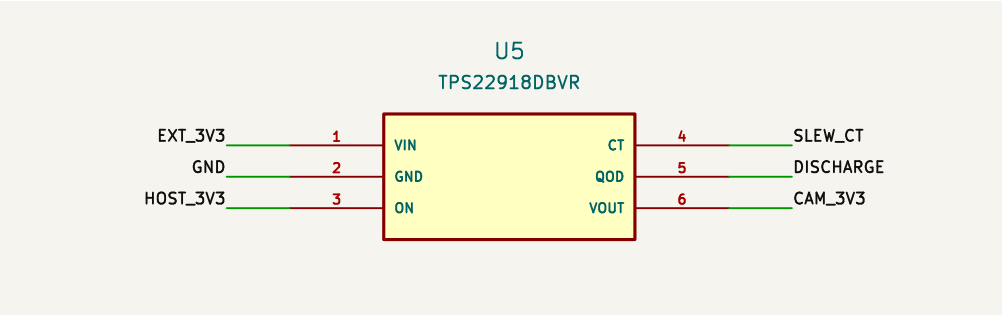
Control and camera power

Selector, control header, load switch and external input

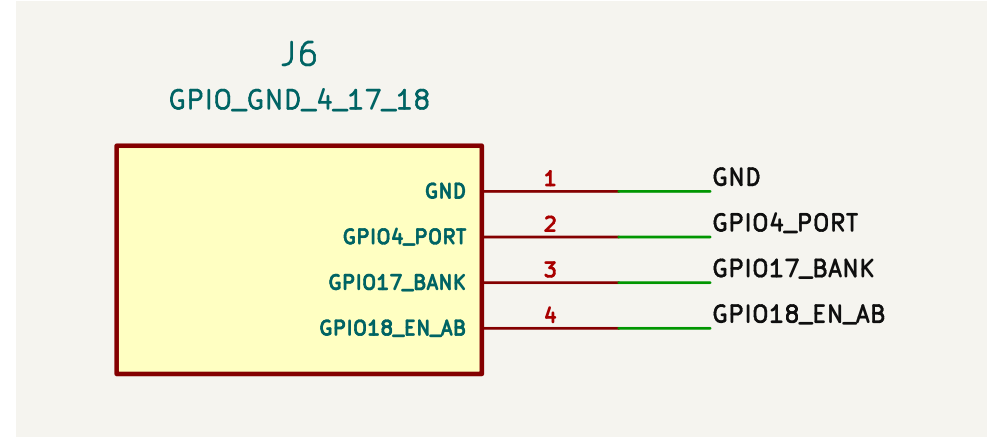
U4 / PCA9544A I2C selector



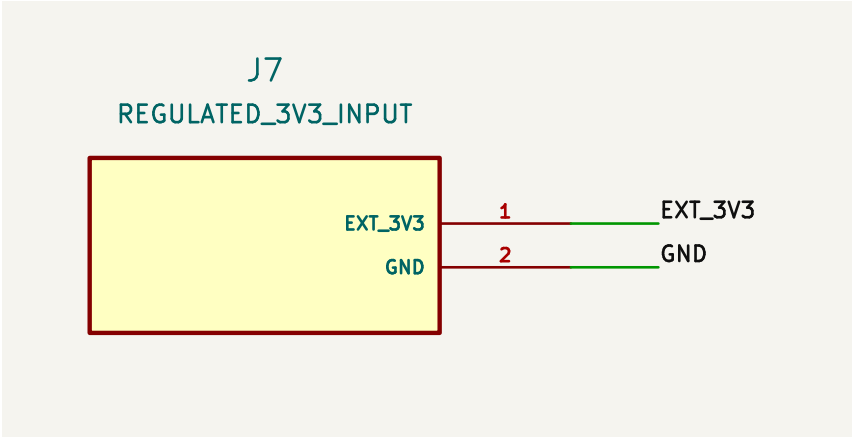
U5 / TPS22918 load switch



J6 / GPIO control



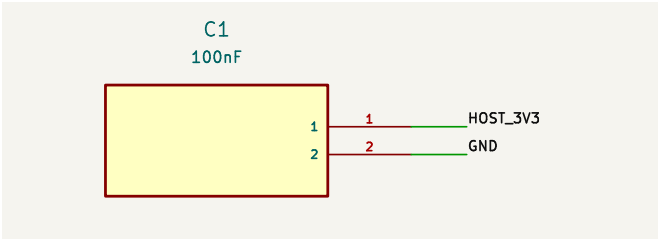
J7 / regulated 3.3 V input



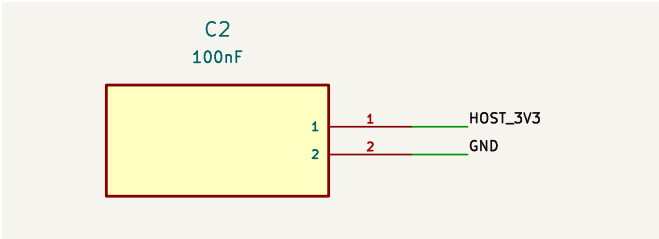
Decoupling and supply markers

Individual source regions; DNP crosses and values are preserved

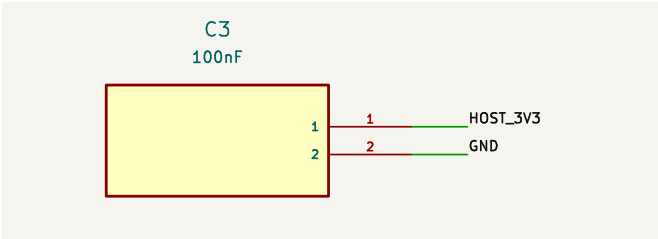
C1



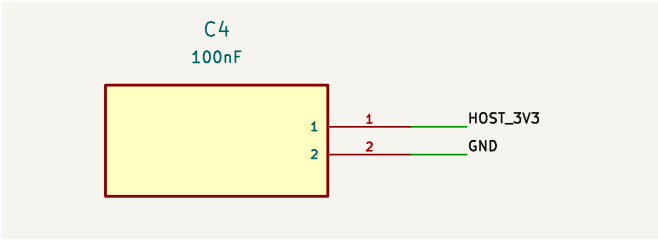
C2



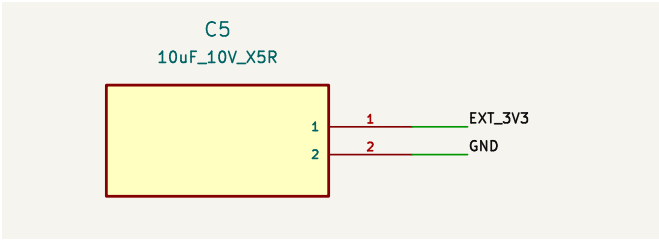
C3



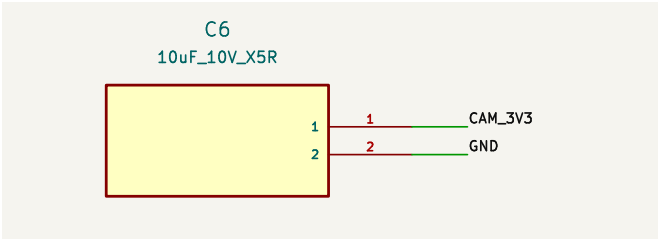
C4



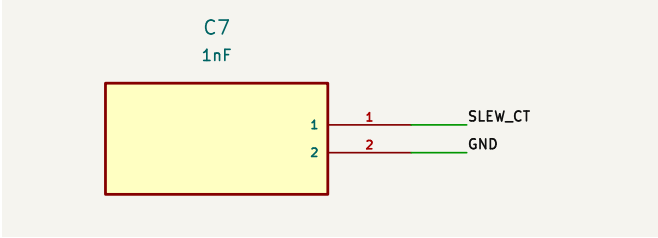
C5



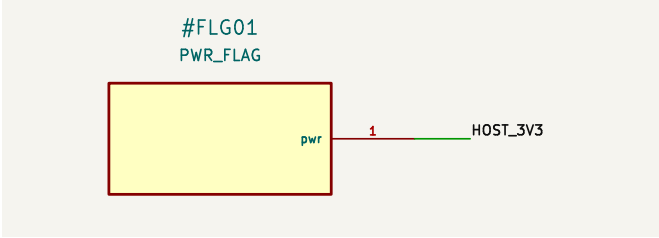
C6



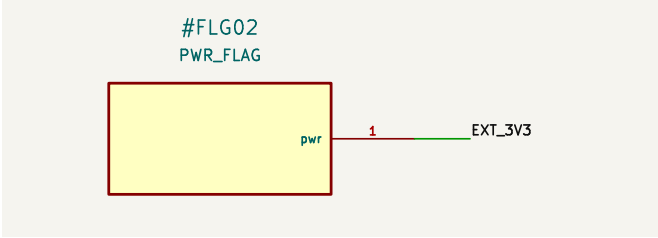
C7



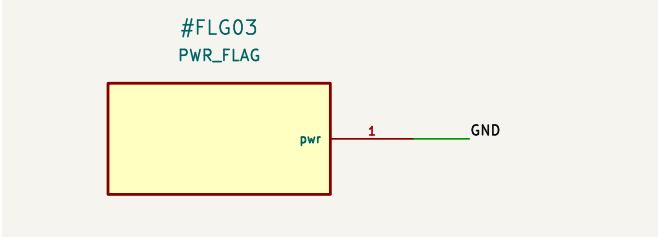
#FLG01



#FLG02



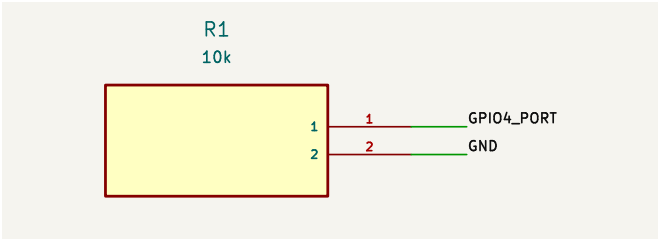
#FLG03



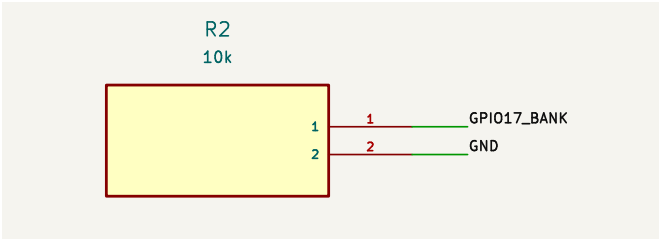
Control bias and host pull-ups

Individual source regions; DNP crosses and values are preserved

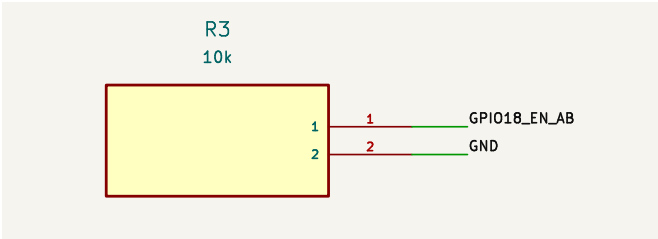
R1



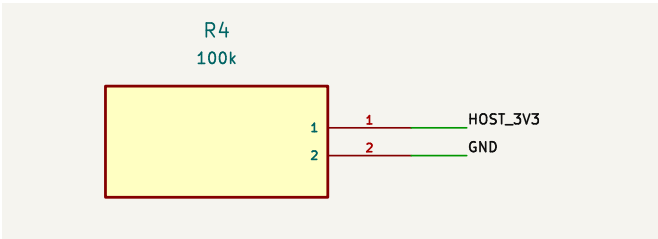
R2



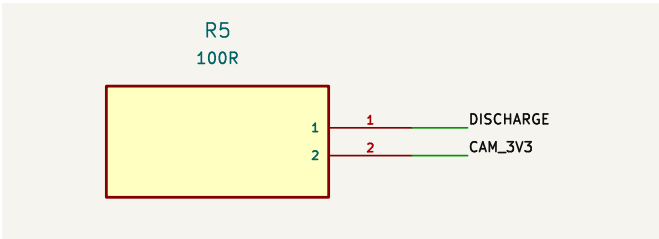
R3



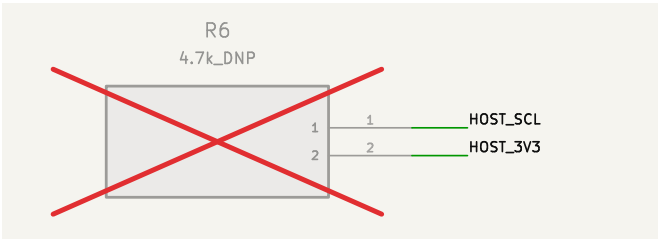
R4



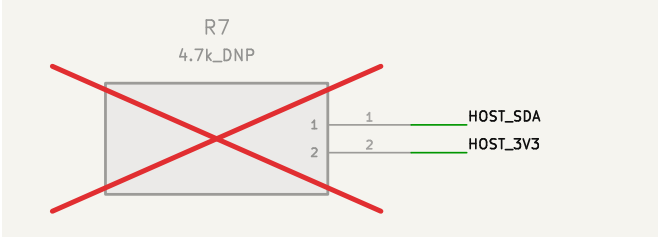
R5



R6



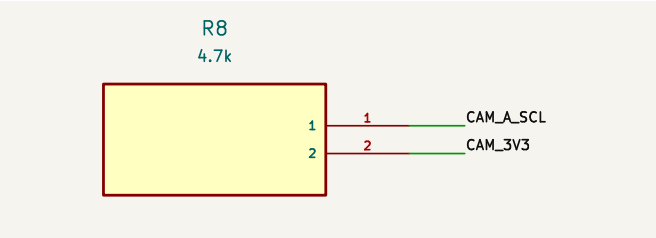
R7



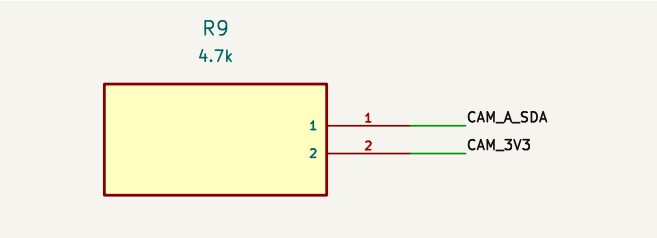
Camera I2C pull-ups

Individual source regions; DNP crosses and values are preserved

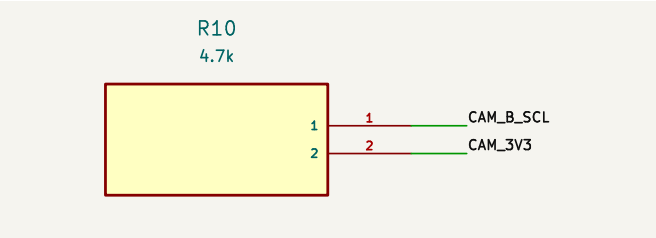
R8



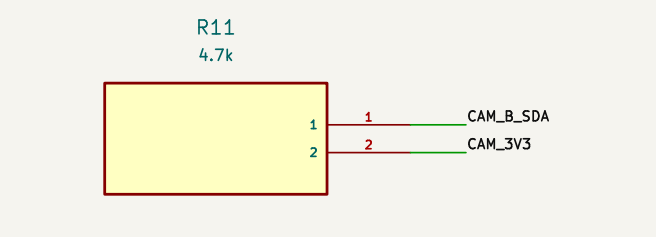
R9



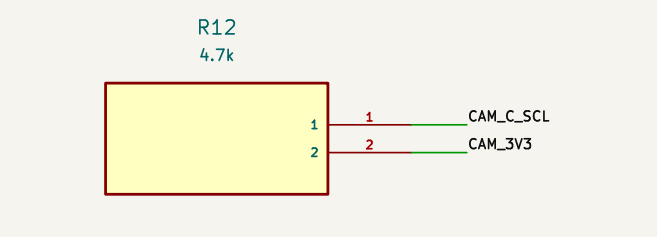
R10



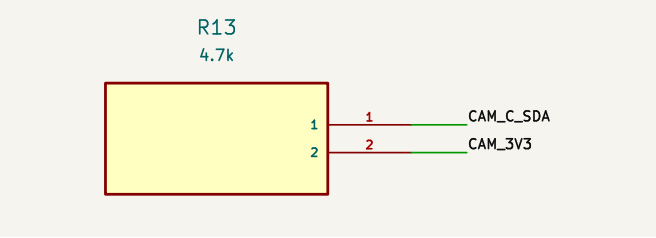
R11



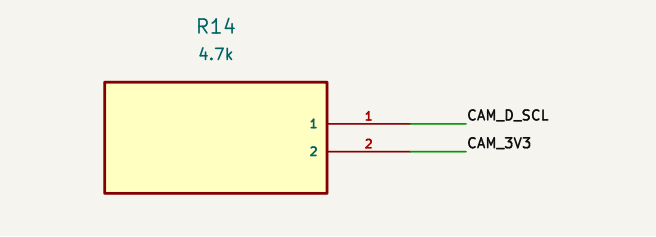
R12



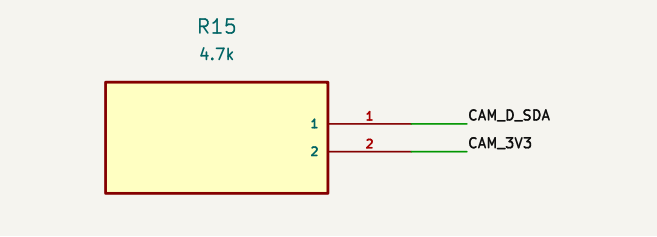
R13



R14



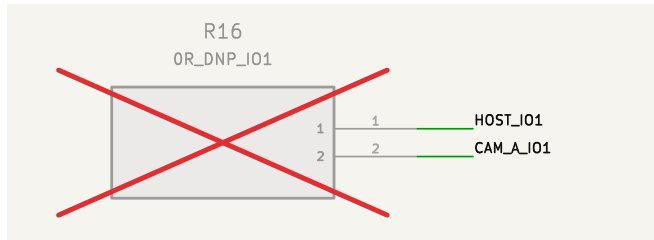
R15



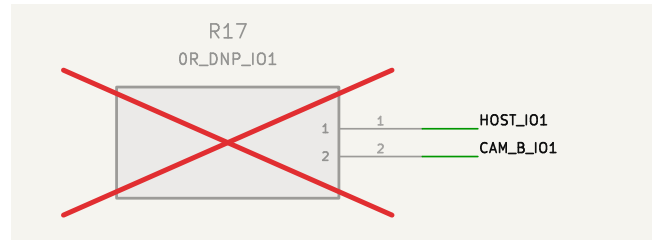
Optional camera IO1 links

Individual source regions; DNP crosses and values are preserved

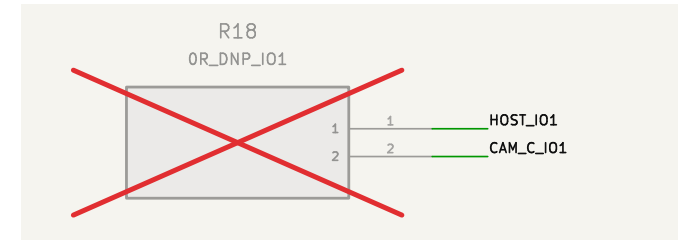
R16



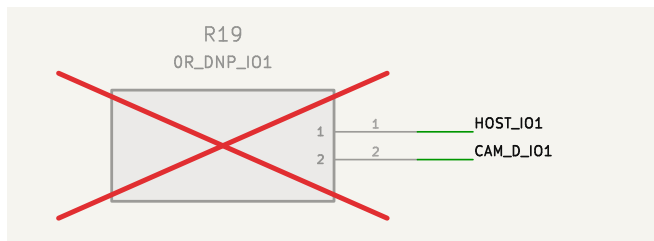
R17



R18



R19



Original design notes

A: GPIO4/18/17 = 0/1/0; B = 1/1/0; C = 0/0/1; D = 1/0/1

Camera IO0 is common for probing. IO1 links R16–R19 are DNP pending camera-specific review.

R6/R7 DNP: host pull-ups normally present. Check parallel module pull-ups before fitting R8–R15.